

Synergy or Star Power: Optimizing Esports Team Rosters

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November 7, 2025

1. Project Description

Recently, I’ve spent quite a lot of time watching professional League of Legends gameplay. During the transfer season, there is a lot of hype about teams creating superstar teams by buying all the best players for every role. Fans and analysts would immediately say things like, “This team is definitely going to win the world championship this year.” However, often, these teams would not be as successful, even during the regular season. I think that, since this is a team sport, sometimes having the best players matters less than how well they work together as a team. To put this theory to the test, I want to conduct a project that directly compares teams built by selecting the highest-rated player for each role—the greedy choice—with teams formed by prioritizing player synergy. This will help determine which approach leads to better team performance.

2. Data

For this project, I will use publicly available competitive League of Legends datasets such as Oracle’s Elixir match database [1]. These datasets include detailed player statistics, match outcomes, laning and objective metrics, champion pools, and historical teammate information. This provides both individual performance features and the relational data needed to estimate player synergy.

3. Methods

I will model roster construction as an optimization problem. Let each player have an individual quality score, and each pair of players have a synergy score based on their historical performance together (and perhaps some other statistics). The goal is to select exactly one player for each of the five roles by maximizing

$$\text{Total score} = \sum (\text{individual quality}) + \sum (\text{pairwise synergy}). \quad (1)$$

This allows me to compare two approaches: a greedy roster that maximizes individual quality, and a synergy-aware roster that optimizes the overall objective. A key part is estimating both individual quality and synergy from player statistics obtained from [1]. The results will demonstrate the extent to which synergy enhances predicted team performance.

4. Expected Results

I expect synergy-aware rosters to outperform greedy lineups. I also expect this to show in match data, with teams that have higher total scores winning more matches than those with lower scores.

5. Practical Implications

A model that incorporates both individual quality and synergy can help teams make more informed roster decisions, especially during the offseason when numerous combinations are possible. This framework can also be applied to other esports or traditional team sports.

References

- [1] Tim Sevenhuysen. Oracle’s elixir: League of legends esports stats. <https://oracleselixir.com/>, 2025. Accessed: November 7, 2025.